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PUBLISHED BY
ELSEVIER

The Journal of Pain, Vol 25, No. 8 (August), 2024: 104500
Available online at www.jpain.org and www.sciencedirect.com

Reduction of Chronic Primary Low Back Pain by Spinal Manipulative Therapy is Accompanied by Decreases in Segmental Mechanical Hyperalgesia and Pain Catastrophizing: A Randomized Placebo-controlled Dual-blind Mixed Experimental Trial

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Abstract: **Chronic primary low back pain (CPLBP) refers to low back pain that persists over 3 months, that cannot be explained by another chronic condition, and that is associated with emotional distress and disability. Previous studies have shown that spinal manipulative therapy (SMT) is effective in relieving CPLBP, but the underlying mechanisms remain elusive. This randomized placebo-controlled dual-blind mixed experimental trial (NCT05162924) aimed to investigate the efficacy of SMT to improve CPLBP and its underlying mechanisms. Ninety-eight individuals with CPLBP and 49 controls were recruited. Individuals with CPLBP received SMT (n = 49) or a control intervention (n = 49), 12 times over 4 weeks. The primary outcomes were CPLBP intensity (0–100 on a numerical rating scale) and disability (Oswestry Disability Index). Secondary outcomes included pressure pain thresholds in 4 body regions, pain catastrophizing, Central Sensitization Inventory, depressive symptoms, and anxiety scores. Individuals with CPLBP showed widespread mechanical hyperalgesia ($P < .001$) and higher scores for all questionnaires ($P < .001$). SMT reduced pain intensity compared with the control intervention (mean difference: −11.7 [95% confidence interval, −11.0 to −12.5], $P = .01$), but not disability ($P = .5$). Similar mild to moderate adverse events were reported in both groups. Mechanical hyperalgesia at the manipulated segment was reduced after SMT compared with the control intervention ($P < .05$). Pain catastrophizing was reduced after SMT compared with the control intervention ($P < .05$), but this effect was not significant after accounting for changes in clinical pain. Although the reduction of segmental mechanical hyperalgesia likely contributes to the clinical benefits of SMT, the role of pain catastrophizing remains to be clarified.**

Perspective: ***This randomized controlled trial found that 12 sessions of SMT yield greater relief of CPLBP than a control intervention. These clinical effects were independent of expectations, and accompanied by an attenuation of hyperalgesia in the targeted segment and a modulation of pain catastrophizing.***

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Key Words: *Chronic nonspecific low back pain, chiropractic, spinal manipulation, placebo, nociplastic pain*

Received September 3, 2023; Received in revised form February 4, 2024; Accepted February 11, 2024
Supplementary data accompanying this article are available online at www.jpain.org and www.sciencedirect.com.
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1526-5900/$36.00
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https://doi.org/10.1016/j.jpain.2024.02.014

Spine pain is the leading cause of disability worldwide[1] and is a personal, economic, and social burden.[2,3] A recent study reported that low back and neck pain were the conditions for which health care spending was the largest, exceeding spending for cardiovascular diseases and cancer.[4] It was also reported that for low back pain (LBP), ineffective low-value care is overused, and effective high-value care is underused.[5] This may worsen LBP and increase health care spending.[5,6] Thus, high-value interventions must be established. Also, identifying factors associated with a positive response to high-value care may improve the management of LBP through personalized interventions.

To classify musculoskeletal pain on mechanistic bases, 3 descriptors were proposed: nociceptive, neuropathic, and nociplastic.[7,8] One advantage of this system is the possibility to classify painful conditions with unclear etiology,[7,9-11] including chronic LBP.[12,13] For cases defined as chronic primary LBP (CPLBP), pain is the primary condition rather than a symptom secondary to a musculoskeletal condition.[12,14] In CPLBP, altered nociceptive processing is presumed to contribute to so-called nociplastic pain more than pathology in specific musculoskeletal tissues.[10] Accordingly, central sensitization likely enhances spinal nociceptive transmission in CPLBP.[11,13,15-17]

In addition to education, reassurance and a progressive return to activities, most clinical practice guidelines recommend spinal manipulative therapy (SMT) for the management of CPLBP[18,19] (reviewed in[20]). Notwithstanding, the efficacy of SMT compared with placebo and its pain-relieving mechanisms remain elusive.[21] These effects are small, not clinically significant, and uncertain, owing to low quality evidence.[22,23] Contextual factors may significantly contribute to the effects of SMT in patients with CPLBP,[24-26] leading to the suggestion that its effects are largely nonspecific.[27,28] However, basic studies suggest that SMT relieves back pain by reducing inflammation and inhibiting nociplastic pain mechanisms, including the enhancement of spinal nociceptive transmission.[21,29-33] Nevertheless, how these results apply to CPLBP patients in clinical settings remains unclear.[28,34]

Based on the proposed mechanisms of SMT and the heterogeneous pathophysiology of chronic LBP, patients presenting with nociplastic LBP may respond differently to SMT depending on individual factors.[21,35,36] Targeting patients with favorable prognostic factors may contribute to the improvement of clinical outcomes.[11,37] However, the reduction of nociplastic pain and associated factors by SMT, and whether these changes contribute to clinical outcomes remain to be clarified.

The aim of the present study was to assess processes underlying nociplastic pain and associated factors in patients with CPLBP treated with SMT to determine how they contribute to CPLBP and its clinical improvement. We hypothesized that the reduction of pain intensity and disability would be greater for SMT compared with a validated placebo intervention, and that measures of nociplastic pain and associated factors at baseline would predict these changes. We also hypothesized that changes in processes underlying nociplastic pain and associated factors contribute to CPLBP improvement.

Methods

This manuscript follows the Consolidated Standards of Reporting Trials guidelines for reporting randomized clinical trials[38] and the Template for Intervention Description and Replication guide for reporting placebo and sham controls (TIDieR–placebo).[39] The latter is reported using the recent update including items from the Recommendations for the Development, Implementation, and Reporting of Control Interventions in Efficacy Trials of Physical, Psychological, and Self-Management Therapies (CoPPS) Statement.[40] The study protocol has been reported elsewhere, before the completion of this study.[41] The study is a randomized placebo-controlled dual-blind mixed experimental trial registered a priori with clinicaltrials.gov (NCT05162924). All experimental procedures conformed to the standards set by the latest revision of the Declaration of Helsinki and the guidelines of the International Association for the Study of Pain. Ethics approval was obtained from the Fundación Jiménez Díaz Clinical Research Ethics Committee (Madrid, Spain) and all participants gave written informed consent, acknowledging their right to withdraw from the experiment without prejudice. The trial comprised 3 parallel arms, 2 that included patients with CPLBP, and 1 that included pain-free healthy controls. The primary outcomes were self-reported pain intensity and disability. The secondary outcomes included other clinical measures as well as psychometric and quantitative sensory testing (QST) variables. These variables were proposed for a mechanism-based classification of patients with chronic musculoskeletal pain, including CPLBP.[42] In the present study, they were used as predictors and moderators of changes in the primary outcomes before and after the intervention. The initial protocol included urine samples, but they could not be included in the present study due to technical issues with the samples.

Participants

Participants were recruited between December 2021 and December 2022 at the Madrid College of Chiropractic teaching Clinic (hereon referred to as "the Clinic"). The Clinic is a primary care facility specialized in spine care. Recruitment was conducted by advertisement in San Lorenzo de El Escorial (Madrid, Spain), social media platforms, and by word of mouth. Men and women between 18 and 70 years of age with symptoms compatible with CPLBP for at least 3 months were considered for inclusion in the study. For the patient groups, screening was conducted using a selection questionnaire administered online or in person, depending on patient preference. When deemed eligible, patients were scheduled for a physical examination at the Clinic to confirm the diagnosis of CPLBP, and to rule out exclusion criteria: evidence of a specific pathology

causing LBP, evidence of a predominant neuropathic origin of LBP, diagnosis of mental health disorders (with the exception of anxiety and depression), pain of comparable or greater intensity affecting another body region, use of corticosteroids, opioids or anticytokine medication, pregnancy, lumbar fusion surgery or recent laminectomy, and chiropractic SMT received within the previous 12 months.

Participants in the healthy control group were recruited using a selection questionnaire and were age- and sex-matched to the SMT group. They were excluded if they presented acute or chronic pain, or a diagnosis of a systemic, inflammatory, neurological, or psychiatric condition. The recruitment of this group allowed obtaining reference values for psychophysical and psychological outcomes from a healthy population, and their fluctuations over time. Comparing these values to those of patients with CPLBP is essential to interpret the results, and is a strength of the study. For example, reference values are useful to show whether patients with CPLBP show hyperalgesia and whether this hyperalgesia is only decreased or completely normalized following SMT. Without reference values from a group of healthy volunteers, it is not possible to conclude that patients show hyperalgesia. This also applies to other dependent variables.

Randomization and Group Allocation

Patients were randomized to 1 of the 2 groups receiving real or placebo SMT (hereon referred to as "SMT" and "placebo," respectively). Using a random number generator, a simple randomization sequence was generated by one of the investigators not involved in data collection to obtain a balanced randomization sequence assigning participants to the SMT or placebo groups in a 1:1 ratio. The patients recruited in the study were allocated either to the SMT or placebo groups according to this randomization sequence. Group allocation was concealed from patients and investigators, except for the investigator that delivered SMT and placebo. The efficacy of blinding was assessed after the first, sixth and twelfth (last) treatment sessions. Patients were asked whether they thought that they received a real chiropractic treatment for back pain (Yes or No). In addition, they were asked to rate their certainty of having received the real treatment on a scale from 0 to 100, where 0 indicates that they were certain of not receiving a real chiropractic treatment and 100 indicates that they were certain of receiving a real chiropractic treatment.[43]

Interventions

The interventions were provided to the patient groups over 4 weeks. Healthy controls did not receive any intervention, but all experimental measures were taken over the same time period. This allowed controlling for nonspecific temporal fluctuations in outcome variables. The same chiropractor (20 years of clinical experience) delivered both SMT and the placebo. The placebo intervention was practiced extensively before the study was initiated. All sessions were provided individually for a duration of approximately 10 minutes, always in the same room, using the same treatment table, with the same ambient conditions. Both interventions were explained verbally to participants with identical instructions before the first session. Once the treatment protocol was initiated, verbal and nonverbal communication was kept to a minimum necessary for both groups.

Spinal Manipulative Therapy

Patients in the SMT group received SMT over 4 weeks at a rate of 3 times per week and 1 day between sessions. This protocol was based on current literature on SMT for CPLBP.[44] In each SMT session, the patient received a high-velocity low-amplitude spinal manipulation targeting the most painful vertebral segment, bilaterally.[41] First, manual palpation of the spine was performed in a prone position to localize the vertebral segment to be manipulated, as determined during the initial examination (most painful segment). Then, right and left lumbar manipulations were performed in a side posture, with a force sufficient to generate joint cavitation (associated with an audible release). If the manipulation did not cause joint cavitation, the procedure was repeated once.

Placebo Intervention

Patients in the placebo group received a validated placebo intervention[43] over 4 weeks at a rate of 3 times per week and 1 day between sessions. Although no SMT was used, manual palpation of the spine was performed in a prone position to localize the most painful vertebral segment, as in the SMT group. With the patient in a side posture, a force was then applied to the gluteal region, bilaterally.[41,43] This simulates SMT, but the force is applied with lower velocity compared with SMT and produces no cavitation. Previous studies on SMT have used this validated placebo intervention.[32,45,46] Patients in the placebo group were offered the possibility to receive SMT at the end of the protocol, following the same treatment procedures as the SMT group. These sessions were documented until the last follow-up.

Outcomes

Outcome variables used in the present study (clinical, questionnaire scores and QST) are based on established methods for the classification of patients based on pain mechanisms.[42,47] Some of these variables were used to assess the efficacy of SMT (primary outcome variables: pain intensity and disability), while others (secondary outcome variables) were used as predictors or moderators of changes in the primary outcome variables produced by SMT. All outcome assessment, including adverse events, were conducted by investigators blinded to group allocation that were not involved in providing care, while the investigator delivering care was blinded to the outcome assessment.

Primary Outcomes

CPLBP intensity and disability were the primary outcomes of this study, with changes between baseline and

4 weeks as the primary endpoint. In order to measure the effects of the 12 sessions at the primary endpoint, the primary outcomes were measured during the initial examination and 48 hours after the twelfth session. For exploratory purposes, primary outcomes were also measured at 2 additional time points (8 and 16 weeks after baseline). Pain intensity was rated using a numerical rating scale displayed on a computer screen, with left and right anchors as 0 (no pain) and 100 (worst pain imaginable).[48] Patients rated the current, maximum, minimum, and average pain intensity in the preceding 7 days (for baseline) and since the last session (for the primary endpoint). Average pain intensity was used for all statistical analyses and other measures were used to provide additional details (descriptive statistics). Disability was assessed using the Oswestry disability index (ODI). The ODI is calculated based on a questionnaire that comprises 10 questions, providing a score from 0 to 50, where "0" indicates no disability and "50" indicates the worst disability.[49]

Secondary Outcomes

Quantitative sensory testing. QST was used to assess changes in pain processing, some of which may be used as surrogates of central sensitization.[50] Pressure pain thresholds (PPTs) are the minimum pressure necessary to elicit pain.[51] They are used to examine mechanical sensitivity in deep musculoskeletal tissues. PPTs were suggested to be one of the most reliable and sensitive methods to assess central sensitization in a CPLBP population.[52-54] Using an algometer (Wagner Force Dial FPX, Greenwich, CT), PPTs were assessed at 4 different body sites by using 2 consecutive measurements at a rate of ~50 kPa/s. Pressure was applied over the erector spinae muscles, 2.5 cm from the spinous process[55] of the most painful lumbar segment, bilaterally. Hypersensitivity at this site in the patient groups compared with controls was used to confirm local mechanical hyperalgesia (hereon referred to as segmental PPTs). To assess heterosegmental spreading of mechanical hyperalgesia, PPTs were assessed 4 segments cranial to the most painful lumbar segment, and compared between the patient groups and controls (hereon referred to as heterosegmental PPTs). To assess the segmental spreading of hyperalgesia, PPTs on the lower limb in the dermatome of the most painful lumbar segment were compared between the patient groups and controls (hereon referred to as dermatomal PPTs). To assess widespread hyperalgesia, PPTs on an area unrelated to the innervation of the back (thenar eminence) were compared between the patient groups and controls (remote PPTs).[50,52] For the control participants, PPTs were assessed at the same locations as for the matched participant in the SMT group. The 4 PPT measures were compared between groups at baseline and after 4 weeks, as secondary outcome variables.

Psychometric assessment. The Pain Catastrophizing Scale (PCS)[56] and the Central Sensitization Inventory (CSI)[57] evaluate psychological constructs that are associated with nociplastic pain.[58-61] The CSI measures symptoms presumably related to central sensitization. A cut-off value of 40 points was proposed to infer central sensitization-like mechanisms.[61] Greater CSI scores are associated with greater pain catastrophizing.[59] Accordingly, it was suggested that pain catastrophizing may contribute to central sensitization and nociplastic pain.[42,60,62,63] Validated versions in Spanish language of these questionnaires were compared between patient groups and controls, at baseline and at 4 weeks.[56,57] The internal consistency and reliability are acceptable for both the PCS (Cronbach's α = .79; intraclass correlation coefficient = .84) and the CSI (Cronbach's α = .87; intraclass correlation coefficient = .91).

In addition, symptoms of depression and anxiety were assessed using the Beck Depression Inventory II (BDI) and the Generalized Anxiety Disorder (GAD) scale.[64,65] Depression and anxiety are CPLBP comorbidities.[66,67] Depression and anxiety are associated with, or may aggravate symptoms suggestive of central sensitization and nociplastic pain.[9,17,68-70] The validated Spanish versions of these questionnaires were filled at baseline and at 4 weeks. Both questionnaires have high internal consistency (Cronbach's α = .89 for the BDI, and .94 for the GAD). The GAD also has excellent reliability (intraclass correlation coefficient = .93).

Clinical examination. Data on CPLBP location, duration and frequency were collected at baseline. Pain duration was quantified in months since the onset of the first episode, while pain frequency was classified as fluctuating or episodic.[71] In addition, CPLBP was classified as proportionate or disproportionate, with a discrete or diffuse anatomical distribution, following criteria defined previously.[13,70] Diffuse pain may be reflective of spreading hyperalgesia, suggestive of nociplastic pain.[9,11,72] The classification of symptoms as proportionate or disproportionate was based on the pattern of pain provocation and aggravation, to rule in or out nociceptive pain versus nociplastic pain (suggestive of central mechanisms).[11]

Expectations of Pain Relief

Expectations of pain relief by SMT were measured in the patient groups at baseline. They were rated using a numerical rating scale with numerical anchors on the left, center and right, for total pain relief (–100), no change (0) and maximum pain increase (+100), respectively. Expectations may contribute to the placebo effect[73] and may predict treatment response in patients with chronic pain.[74] This measure was used to assess the potential contribution of the placebo response to changes in the primary outcome variables.[73]

Adverse Events

Adverse reactions following each SMT or placebo sessions were reported by patients using an online questionnaire that was completed at the beginning of the following session. The questionnaire requested participants to describe the type of adverse event, the time of onset, the duration, and the severity of the

events. Adverse events were categorized into 4 main categories: muscle stiffness, increased pain, radiating discomfort, and others.[75] Categorical scales were used to report the onset (immediate, up to 24 hours, or > 24 hours), duration (minutes, < 24 hours, 24–48 hours, or > 48 hours), and intensity (very mild to very severe) of these events. Participants were instructed to inform the chiropractor providing SMT in case of severe adverse events, or repeated moderate events. All reported adverse events were monitored by an investigator not involved in clinical care, who also informed the investigator delivering SMT when the patient reported a 30-point increase in pain intensity or moderate/severe adverse events during at least 2 consecutive visits. In this case, the events and the possibility to withdraw from treatment were discussed with patients.

Sample Size Calculation

The present study is a relatively short intervention (1 month) for a chronic condition. Thus, it was powered to detect small to medium effects ($f = .175$). For 2 groups and 2 repeated measures (baseline and 4 weeks) with an effect size of $f = .175$, an alpha of .05, and a power of .8, a sample of 34 patients per group (total of 68) was required to detect statistically significant changes in clinical pain intensity and disability. Based on this calculation and an attrition rate of 10%, 40 participants are sufficient to detect significant effects on the primary outcome variables. However, one objective of the study was to examine if secondary outcome variables predict primary outcome variables. To detect significant effects in a multiple regression model, it is recommended to plan ten sample elements per predictor variable.[76] We were interested in 5 predictor variables, including segmental PPTs, PCS scores, CSI scores, expectations of pain relief, and urinary concentrations of TNF-α (Tumor Necrosis Factor alpha; not used because of technical issues, but planned before the study). Accordingly, a sample size of 50 participants per group was required. Thus, we planned to recruit a final sample of 50 participants per group. This meets the requirements for both the primary and secondary analyses.

Statistical Analysis

All analyses were performed with JASP v0.16.4 (University of Amsterdam, Amsterdam, Netherlands), Jamovi v2.3.21 (University of Newcastle, Newcastle, Australia), and R Studio v2022.7.1.554 (RStudio Inc., Boston, MA). The Kolmogorov-Smirnov test was used to assess the normality of distributions. The Levene's test was used to assess the homogeneity of variance. When the assumptions of normality and/or homogeneity of variance were not met, robust statistical tests were conducted using the WRS2 package (RStudio Inc., Boston, MA).[77] Robust post hoc tests were carried out with the *mcp2atm* function of the same package. To examine the efficacy of SMT, changes in pain intensity and disability (primary endpoint compared to baseline) were compared between groups (SMT vs placebo) using mixed analyses of variance (ANOVA), with sex as a categorical variable, to report potential sex differences. Both intention-to-treat and per-protocol analyses were performed. Changes in pain and disability were also assessed at 8 and 16 weeks after initiating the treatment for exploratory purposes, with separate mixed ANOVAs. All significant effects were decomposed using robust planned contrasts. To control for a different number of SMT sessions received after the primary endpoint, the number of sessions was included as a covariate. Baseline scores of secondary outcomes were also introduced separately as covariates to ensure that the analyses of primary outcomes accounted for any baseline difference. No dependent variable at baseline affected the results and significant effects remained (see Supplementary Table 1).

To confirm that higher values were observed for variables related to nociplastic pain in patients with CPLBP compared with controls, these values were compared between groups (3) and sexes (2) using 2-way ANOVAs for baseline PPTs at each location. The same approach was used for psychological factors, with PCS, CSI, BDI-II, and GAD scores as dependent variables. In addition, 2-way ANOVAs were used to examine changes (delta values) between the 3 groups and sexes. This served to rule out nonspecific temporal effects, since healthy controls received no intervention. To test a priori hypotheses, significant effects were decomposed using robust planned contrasts. Adjusted *P*-values were calculated with a Bonferroni correction for multiple comparisons.

To examine which variables predicted SMT efficacy, differences in pain intensity and disability between baseline and the primary endpoint (delta values) were used as the dependent variables in separate multiple regression models, for which the estimates were obtained using bootstrapping.[78] The Durbin-Watson test was conducted to assess the independence of observations, and multicollinearity was tested using the variance inflation factor. Baseline PPTs, PCS, and CSI questionnaire scores, and expectations of pain relief were entered as predictors. These analyses were conducted for the entire patient cohort first, and then individually for the SMT and placebo groups. To determine the contribution of changes in variables used as surrogates of nociplastic pain to clinical efficacy, the General Linear Model (GLM) was used with changes in PPTs, PCS and CSI scores and expectations introduced separately in different models as moderators. To evaluate their influence on group differences in primary outcomes, the group was introduced as the fixed factor, and changes in each of the primary outcomes as dependent variables, in separate models for pain intensity and disability.

Finally, expectations of pain relief were compared between groups using independent t-tests. The degree of certainty of having received SMT was compared between groups over time (3 measurements) using mixed ANOVAs, with sex as a categorical variable. Greenhouse-Geisser adjustments were used as needed. To control for the potential influence of expectation and certainty, values were also included as covariates in the analyses of primary outcomes that yielded significant results.[73] Mann-Whitney *U* tests were used to compare the total number of adverse events per participant between treatment groups. The type, onset, duration, and severity of these adverse events were compared between the SMT and the placebo group using Chi-square tests.

Results

Baseline Demographic Characteristics

For the 2 patient groups, a total of 155 individuals were screened for eligibility (see Consolidated Standards of Reporting Trials diagram in Fig 1). Fifty-three were excluded and 102 were randomized to receive the interventions. After the physical examination, 4 randomized participants were excluded (2 presented no pain at all in the previous 7 days, 1 was taking opioids, and 1 presented signs and symptoms of neuropathic LBP). Thus, a total of 98 patients were included in the study, 49 patients per group, with similar baseline characteristics (see Table 1).

A total of 87 healthy volunteers were screened for eligibility. After excluding 38 candidates (20 presented signs and symptoms that were compatible with the exclusion criteria and 18 did not respond or declined to participate), 49 healthy volunteers were included in the control group. Controls were age/sex-matched to patients in the SMT group. Baseline demographic characteristics of all participants are presented in Table 1.

Mechanical Pain Sensitivity and Psychological Characteristics at Baseline

Baseline PPTs were compared between the 3 groups and sexes using a 2-way ANOVA for each of the 4 locations (see Table 2 and Fig. 2A–2D). PPTs were significantly different between groups at the 4 regions (segmental: $F_{2,141} = 91.0$, $P < .001$, $\eta_p^2 = .56$ [Fig 2A]; heterosegmental: $F_{2,141} = 50.5$, $P < .001$, $\eta_p^2 = .42$ [Fig 2B]; dermatomal: $F_{2,141} = 44.9$, $P < .001$, $\eta_p^2 = .39$ [Fig 2C]; remote: $F_{2,141} = 25.4$, $P < .001$, $\eta_p^2 = .26$; Fig 2D), but these effects were not significantly different between sexes (all P's > .1). Bonferroni-corrected planned contrasts revealed that all PPT's were significantly lower for the SMT group compared with controls and for the placebo group compared with controls (all corrected P's < .01). However, they were not significantly different between the SMT and placebo groups (all corrected P's > .4). This indicates that patients with CPLBP from both groups showed widespread hyperalgesia, consistent with nociplastic pain. Besides, PPTs were lower in females compared with males for the 3 groups combined, for the 4 locations (all corrected P's < .01). This suggests that mechanical pain sensitivity is greater in women, regardless of chronic pain.

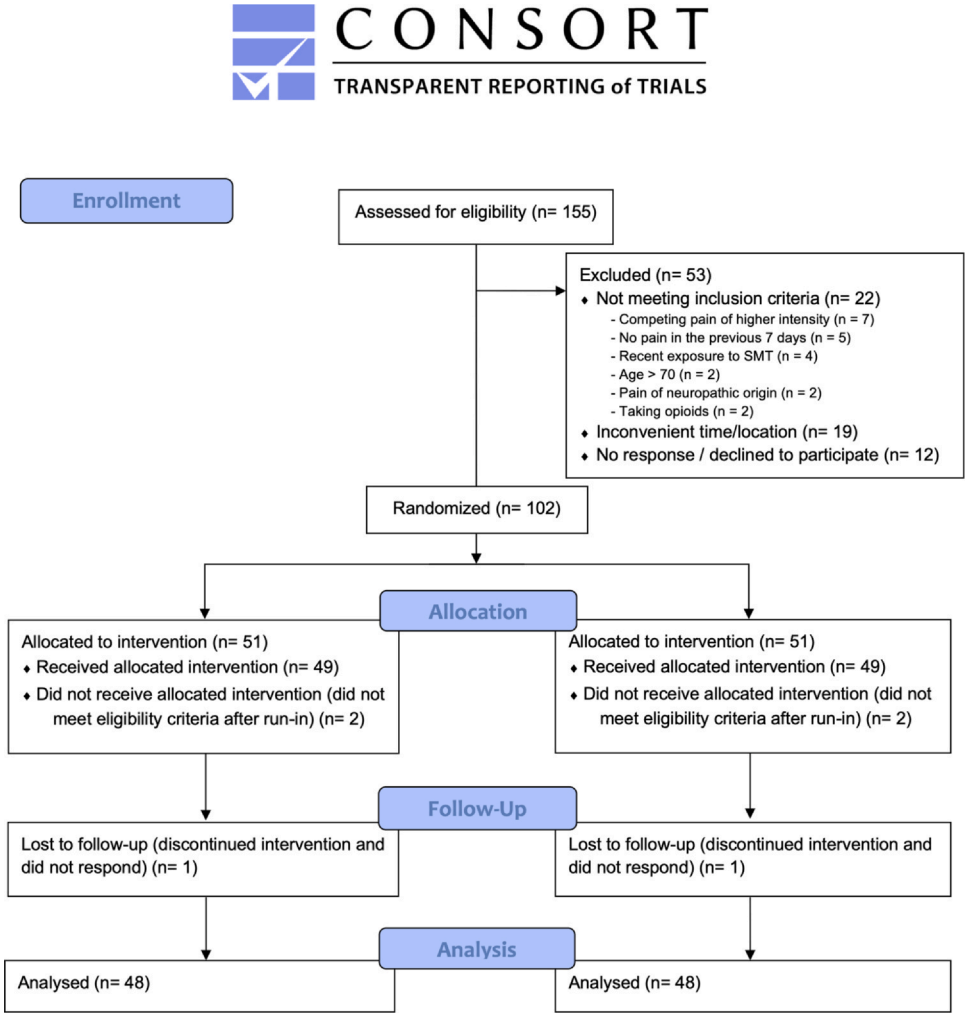


Figure 1. Consolidated Standards of Reporting Trials (CONSORT) diagram.

Table 1. Baseline Demographic Characteristics of Participants

BASELINE CHARACTERISTIC	*PLACEBO GROUP*	*SMT GROUP*	*CONTROL GROUP*
Participants, n	49	49	49
Sex, n (%)			
Women	32 (65.3)	31 (63.3)	31 (63.3)
Men	17 (34.7)	18 (36.7)	18 (36.7)
Age, years (Standard Deviation: SD)	48.3 (9.5)	48.3 (13.0)	48.3 (13.2)
Level of education, n (%)			
High School	23 (46.9)	24 (49.0)	13 (26.5)
University	26 (53.1)	25 (51.0)	36 (73.5)
Smoking (Yes), n (%)	13 (26.5)	16 (32.6)	10 (20.4)
Average daily sleep time, hours (SD)	6.9 (1.1)	6.7 (1.3)	7.1 (1.0)

Table 2. Mechanical Pain Sensitivity and Psychological Characteristics at Baseline

PAIN PHENOTYPING CHARACTERISTIC	*PLACEBO GROUP*	*SMT GROUP*	*CONTROL GROUP*
Pressure pain thresholds, kPa (95% CI)			
Segmental	314 (233–396)***	294 (240–347)***	654 (588–720)
Heterosegmental	359 (278–440)***	364 (303–426)***	670 (598–742)
Dermatomal	310 (238–382)***	271 (228–313)***	529 (462–597)
Remote	300 (247–353)**	273 (227–318)***	418 (382–454)
Questionnaire scores (95% CI)			
Central Sensitization Inventory (0–100)	28.8 (24.8–32.9)***	30.3 (25.9–34.6)***	16.5 (14.0–19.0)
Pain Catastrophizing Scale (0–52)	23.3 (19.9–26.7)***	21.8 (18.9–24.7)**	5.8 (3.5–8.1)
Beck Depression Inventory II (0–63)	7.0 (5.1–8.9)*	8.7 (6.4–11.0)*	3.2 (2.1–4.4)
Generalized Anxiety Disorder scale (0–21)	4.4 (3.2–5.7)*	4.5 (3.2–5.9)**	1.6 (1.0–2.2)

*$P < .05$ compared with controls.
**$P < .01$ compared with controls.
***$P < .001$ compared with controls.

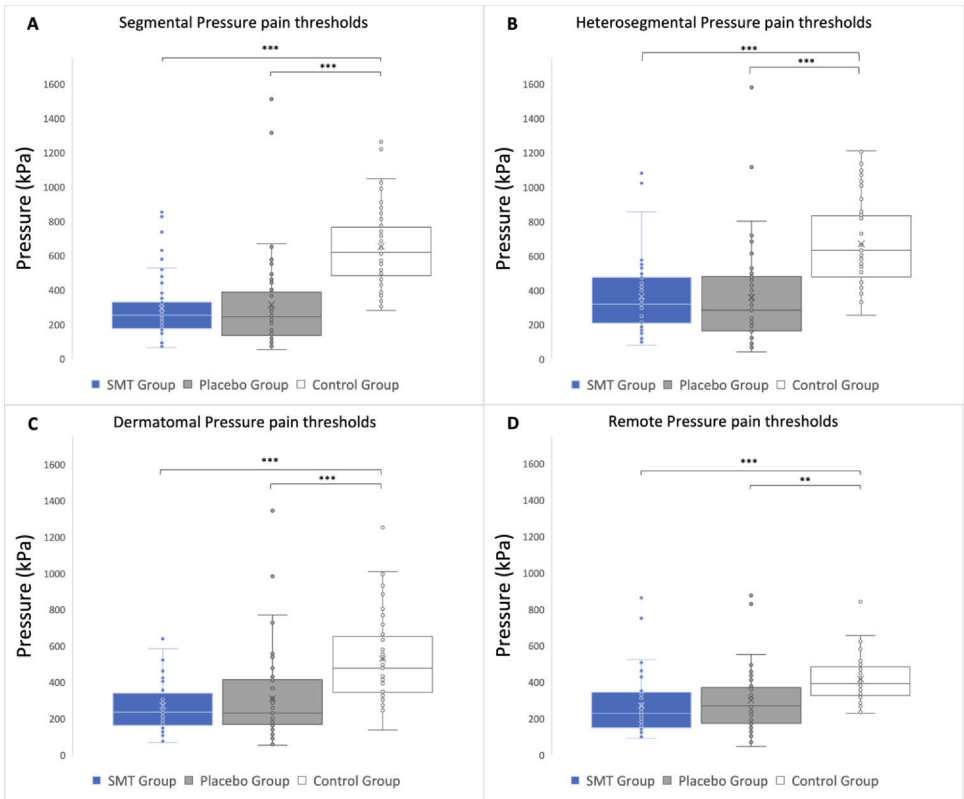


Figure 2. Baseline PPTs. Boxplots showing the mean, median, and interquartile ranges of baseline pressure pain thresholds for (**A**) segmental, (**B**) heterosegmental, (**C**) dermatomal, and (**D**) remote areas, for participants in the SMT, placebo, and control groups. PPTs were significantly lower for both patient groups compared with the control group. **$P < .01$, ***$P < .001$ between groups.

Baseline scores for questionnaires (CSI, PCS, BDI, and GAD) were compared between the 3 groups and sexes using 2-way ANOVAs (see Table 2). The scores from the 4 questionnaires were significantly different between groups (CSI: $F_{2,141} = 32.5$, $P < .001$, $\eta_p^2 = .31$; PCS: $F_{2,141} = 87.1$, $P < .001$, $\eta_p^2 = .55$; BDI: $F_{2,141} = 12.2$, $P = .005$, $\eta_p^2 = .15$; GAD: $F_{2,141} = 18.6$, $P < .001$, $\eta_p^2 = .21$). These effects were not significantly different between sexes (all P's $> .05$). Bonferroni-corrected planned contrasts revealed significantly higher scores for all questionnaires for the SMT group and the placebo groups compared with the control group (all corrected P's $< .05$). Moreover, no significant difference was observed between the 2 patient groups (all corrected P's $> .8$). Besides, no significant difference was observed between sexes for any questionnaire (all P's $> .05$). These results indicate that patients with CPLBP from both groups show a profile consistent with nociplastic pain.

Efficacy of SMT

Out of 98 patients included in the clinical trial, 96 completed the protocol. Two patients discontinued treatment because they could not commit to complete the protocol. The 96 remaining participants adhered to the protocol as planned (see Table 3 for baseline clinical characteristics). This resulted in 2.04% of missing data. Little's MCAR (Missing Completely At Random) test showed a χ^2 value of 9.8 ($P = 1.0$), indicating that the values were missing completely at random. Both the per-protocol and intention-to-treat analyses using the expectation-maximization algorithm for the missing primary endpoint measurements were conducted.[79,80] They yielded equivalent results and conclusions. Thus, only the per-protocol analyses are reported here for clarity.

SMT efficacy was assessed by comparing average pain intensity and disability between baseline and the primary endpoint and between groups and sexes using mixed ANOVAs (see Fig 3). Average pain intensity at baseline was 47.5 (95% confidence interval [CI]: 42.5–52.5) for the SMT group, and 45.8 (95% CI: 40.1–51.5) for the placebo group. At the primary endpoint, average pain intensity was reduced to 16.6 (95% CI: 12.5–20.7) for the SMT group, and to 27.9 (95% CI: 21.1–34.8) for the placebo group. Baseline disability scores were 9.8 (95% CI: 7.9–11.7) for the SMT group, and 9.8 (95% CI: 8.0–11.5) for the placebo group. At the primary endpoint, disability scores were 4.6 (95% CI: 3.3–5.9) for the SMT group, and 5.6 (95% CI: 4.0–7.1) for the placebo group. Average pain intensity was significantly different between groups over time ($F_{1,92} = 6.9$, $P = .01$, $\eta_p^2 = .07$; mean difference: −11.7; 95% CI: −19.7 to −3.7), indicating that pain relief produced by SMT was greater compared with placebo (see Fig 3). This effect was not significantly different between sexes ($F_{1,92} = 2.4$, $P = .12$, $\eta_p^2 = .03$). For disability, the between-group difference over time of −.9 points (95% CI: −3.0 to 1.3) was not statistically significant ($F_{1,92} = .6$, $P = .5$, $\eta_p^2 = .01$; see Fig 4), indicating that SMT did not reduce disability significantly compared with placebo. Also, the Group × Sex × Time interaction was not statistically significant ($F_{1,92} = .1$, $P = .8$, $\eta_p^2 = .001$).

Patients were contacted at 8 and 16 weeks (4 and 12 weeks after the end of the protocol) to examine if the clinical improvement persisted. At the 8-week follow-up, clinical pain intensity remained lower for SMT compared with placebo ($F_{1,87} = 8.7$, $P = .004$, $\eta_p^2 = .09$; mean difference: −13.2; 95% CI: −22.3 to −4.1; n = 44 vs 47), but the Group × Sex × Time interaction was not statistically significant ($F_{1,87} = .04$, $P = .8$, $\eta_p^2 < .001$). When controlling for the number of SMT sessions after the end of the protocol, the effect remained comparable ($F_{1,86} = 9.9$, $P = .002$, $\eta_p^2 = .10$). For disability, no significant difference was observed between groups ($F_{1,87} = .8$, $P = .4$, $\eta_p^2 = .01$; mean difference: −.7; 95% CI: −3.2 to −1.8; n = 44 vs 47), and the Group × Sex × Time interaction was not statistically significant ($F_{1,87} = 1.3$, $P = .2$, $\eta_p^2 = .01$). Controlling for the number of SMT sessions after the end of the protocol did not change the results ($F_{1,86} = 1.1$, $P = .3$, $\eta_p^2 = .01$).

Table 3. Clinical Characteristics of Patient Groups at Baseline

CLINICAL CHARACTERISTIC	PLACEBO GROUP	SMT GROUP
Pain duration, months (95% CI)	110 (71–148)	118 (84–151)
Average pain intensity, 0 to 100 (95% CI)	45.8 (40.1–51.5)	47.5 (42.5–52.5)
Current pain intensity, 0 to 100 (95% CI)	36.7 (32.2–47.2)	40.6 (34.3–46.7)
Maximum pain intensity, 0 to 100 (95% CI)	59.8 (53.5–66.1)	62.9 (57.3–68.5)
Minimum pain intensity, 0 to 100 (95% CI)	27.2 (21.1–33.3)	26.6 (20.4–32.7)
Oswestry disability index score, 0 to 50 (95% CI)	9.8 (8.0–11.5)	9.8 (7.9–11.7)
Pain extent, number of pixels $\times 10^3$ (95% CI)	32.9 (19.8–45.9)	33.7 (23.1–44.3)
Expectations of relief, −100/+100 (95% CI)	−68.7 (−76.5 to −60.9)	−68.8 (−75.6 to −62.0)
Pain characteristics, n (%)		
Fluctuating	36 (73.5)	40 (81.6)
Episodic	13 (26.5)	9 (18.4)
Diffuse location (Yes)	8 (16.3)	6 (12.2)
Disproportionate (Yes)	13 (26.5)	9 (18.4)
Lower extremity pain (Yes), n (%)	16 (32.6)	21 (42.9)
Taking pain medication (Yes), n (%)	18 (36.7)	22 (44.9)
Chronic comorbidities (Yes), n (%)	19 (36.7)	24 (49.0)

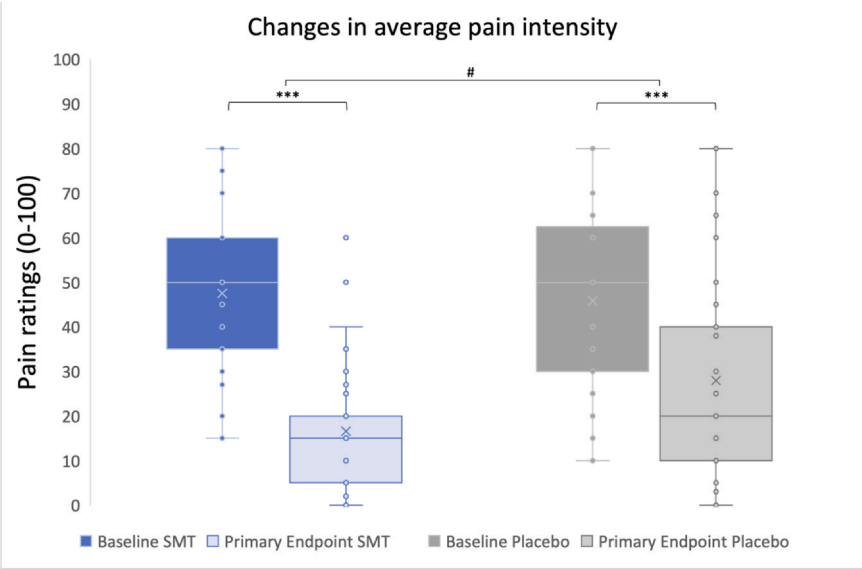


Figure 3. Average pain intensity. Boxplots showing mean, median, and interquartile ranges of average pain intensity ratings (0–100) at baseline and at the primary endpoint (4 weeks), for the SMT and placebo groups. Pain ratings were significantly lower at the primary endpoint compared with baseline for both groups, but the reduction was significantly greater for the SMT group compared with placebo group. * ** $P < .001$ within group, $^{\#\#}P < .01$ between groups.

At the 16-week follow-up, pain intensity remained significantly different between groups ($F_{1,83} = 6.1$, $P = .02$, $\eta_p^2 = .07$; mean difference: –12.4; 95% CI: –22.2 to –2.6; n = 42 vs 45, respectively), but the Group × Sex × Time interaction was not statistically significant ($F_{1,83} = .1$, $P = .7$, $\eta_p^2 = .00$). When controlling for the number of SMT sessions after the end of the protocol, the effect remained unchanged ($F_{1,82} = 19.1$, $P < .001$, $\eta_p^2 = .19$). For disability, no significant difference was observed between groups ($F_{1,83} = 1.6$, $P = .2$, $\eta_p^2 = .02$; mean difference: –.8; 95% CI: –3.4 to 1.8; n = 42 vs 45, respectively), and the Group × Sex × Time interaction was not statistically significant ($F_{1,83} = 2.8$, $P = .1$, $\eta_p^2 = .03$). Controlling for the number of SMT sessions after the end of the protocol did not change the results ($F_{1,82} = 3.5$, $P = .06$, $\eta_p^2 = .04$).

Changes in Mechanical Pain Sensitivity and Psychological Factors

Changes (delta values) in mechanical pain sensitivity (PPTs) and scores of the 4 questionnaires (CSI, PCS, BDI, and GAD) were compared between the SMT and placebo groups and sexes using 2-way ANOVAs (see Table 4).

Changes in segmental PPTs were significantly different between groups (main effect: $F_{2,139} = 4.7$, $P = .01$, $\eta_p^2 = .06$; see Fig 5), but the Group × Sex interaction was not statistically significant ($F_{2,139} = .2$, $P = .8$, $\eta_p^2 = .003$). Segmental

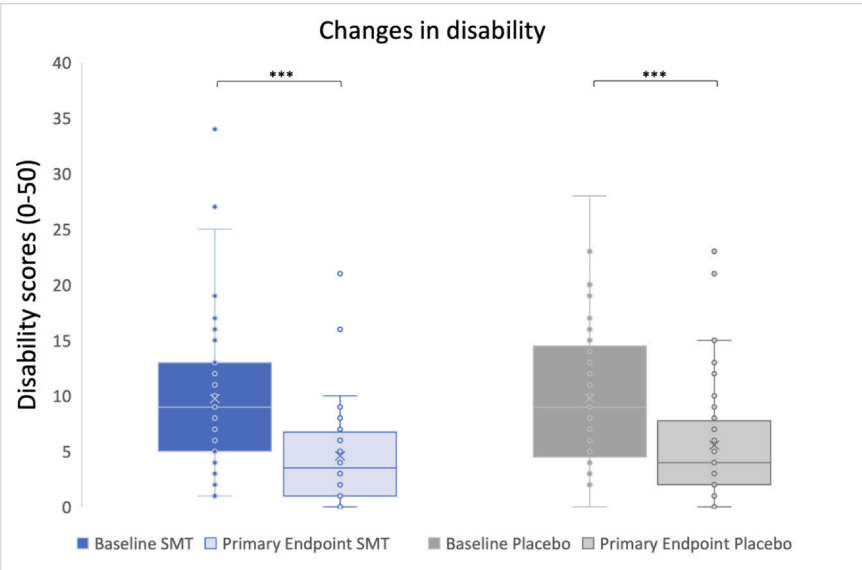


Figure 4. Disability. Boxplots showing the mean, median and interquartile ranges of disability scores (0–50) at baseline and at the primary endpoint (4 weeks), for the SMT and placebo groups. Disability was significantly lower at the primary endpoint compared with baseline for both groups, and the reduction was not significantly different between groups. * **$P < .001$ within group.

Table 4. Changes in Primary and Secondary Outcome Variables (Mean ± Standard Deviation)

OUTCOME VARIABLES	BASELINE	PRIMARY ENDPOINT	8-WEEK FOLLOW-UP	16-WEEK FOLLOW-UP
Pain intensity ratings (0–100)				
SMT Group	47.5 (17.3)	16.6 (14.1)***,†	21.2 (17.6)***,††	19.6 (17.7)***,†
Placebo Group	45.8 (19.9)	27.9 (23.7)***,†	35.1 (25.1)**,††	30.4 (25.1)***,†
Oswestry disability index scores (0–50)				
SMT Group	9.8 (6.6)	4.6 (4.3)***	4.1 (4.4)***	3.9 (4.7)***
Placebo Group	9.8 (6.1)	5.6 (5.3)***	5.4 (5.3)***	4.6 (4.8)***
Segmental pressure pain thresholds (kPa)				
SMT Group	293.5 (184.6)	357.9 (204.4)*,†	-	-
Placebo Group	314.3 (283.2)	309.0 (253.3)†	-	-
Control Group	653.7 (230.0)	653.8 (210.1)	-	-
Heterosegmental pressure pain thresholds (kPa)				
SMT Group	364.5 (214.9)	403.1 (214.3)	-	-
Placebo Group	359.3 (281.7)	358.9 (302.6)	-	-
Control Group	670.0 (251.3)	678.0 (244.4)	-	-
Dermatomal pressure pain thresholds (kPa)				
SMT Group	270.9 (147.7)	302.3 (143.8)	-	-
Placebo Group	309.8 (250.3)	327.5 (272.5)	-	-
Control Group	529.2 (235.4)	533.3 (230.5)	-	-
Remote pressure pain thresholds (kPa)				
SMT Group	272.6 (158.9)	287.2 (140.1)	-	-
Placebo Group	300.2 (184.6)	308.9 (184.4)	-	-
Control Group	418.1 (126.5)	438.6 (144.0)	-	-
Central Sensitization Inventory score (0–100)				
SMT Group	30.3 (15.0)	24.1 (13.0)***	-	-
Placebo Group	28.8 (14.1)	24.7 (12.5)***	-	-
Control Group	16.5 (8.8)	14.5 (7.7)	-	-
Pain Catastrophizing Scale score (0–52)				
SMT Group	21.8 (10.0)	13.7 (8.9)***,††,ÞÞÞ	-	-
Placebo Group	23.3 (11.7)	18.2 (11.0)***,††	-	-
Control Group	5.8 (7.9)	6.1 (7.4)ÞÞÞ	-	-
Beck Depression Inventory II score (0–63)				
SMT Group	8.7 (7.9)	5.8 (6.7)***	-	-
Placebo Group	7.0 (6.5)	4.7 (6.0)***	-	-
Control Group	3.2 (4.0)	2.6 (3.7)*	-	-
Generalized Anxiety Disorder score (0–21)				
SMT Group	4.5 (4.8)	3.2 (4.3)	-	-
Placebo Group	4.4 (4.3)	2.8 (3.7)***	-	-
Control Group	1.6 (2.2)	1.4 (2.4)	-	-

*Significant compared with baseline ($P < .05$).
**Significant compared with baseline ($P < .01$).
***Significant compared with baseline ($P < .001$).
†Significant compared with baseline between the SMT and placebo groups ($P < .05$).
††Significant compared with baseline between the SMT and placebo groups ($P < .01$).
ÞÞÞSignificant compared with baseline between the SMT and control groups ($P < .001$).

PPTs were increased by 61.8 kPa (95% CI: 25.3–98.4) in the SMT group, were decreased by 9.7 kPa (95% CI: –47.4 to 28.1) in the placebo group, and were increased by .05 kPa (95% CI: –31.8 to 31.9) in the control group. Bonferroni-corrected planned contrasts revealed that segmental PPTs increased significantly in the SMT group compared with the placebo group (corrected $P = .016$, Cohen's $d = .51$), and marginally compared with the control group (corrected $P = .051$, $d = .60$).

Changes in heterosegmental, dermatomal, and remote PPTs were not significantly different between groups (main effect: $F_{2,139} = 1.7$, $P = .4$, $\eta_p^2 = .02$; $F_{2,139} = 1.8$, $P = .4$, $\eta_p^2 = .02$; $F_{2,139} = .6$, $P = .8$, $\eta_p^2 = .01$, respectively). Moreover, the Group × Sex interaction was not statistically significant (heterosegmental: $F_{2,139} = 2.6$, $P = .3$, $\eta_p^2 = .04$; dermatomal: $F_{2,139} = 3.1$, $P = .2$, $\eta_p^2 = .04$; remote: $F_{2,139} = .8$, $P = .7$, $\eta_p^2 = .01$).

In summary, these results indicate that segmental, but not heterosegmental, dermatomal, or remote mechanical pain sensitivity was decreased in the SMT group compared with the placebo group.

Regarding questionnaires, changes in CSI scores were not significantly different between groups (main effect: $F_{2,139} = 3.3$, $P = .2$, $\eta_p^2 = .05$), and the Group × Sex interaction was not statistically significant ($F_{2,139} = 1.7$, $P = .4$, $\eta_p^2 = .02$).

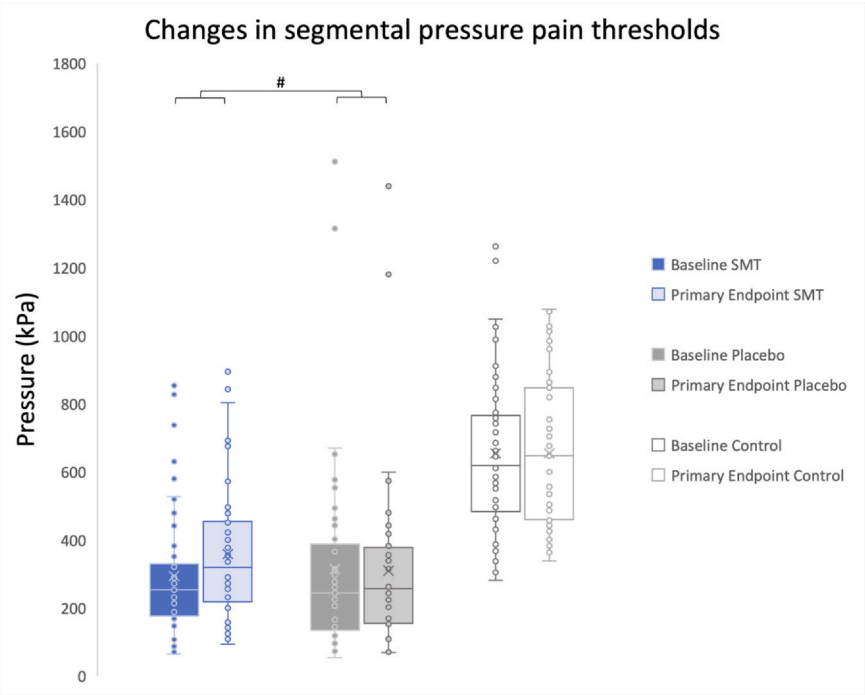


Figure 5. Changes in PPTs. Boxplots showing the mean, median, and interquartile ranges of segmental pressure pain thresholds at baseline and at the primary endpoint (4 weeks), for participants in the SMT, placebo, and control groups. The increase of PPTs over time (indicating decreased pain sensitivity) was significantly greater in the SMT group compared with the placebo group. $^{\#}P < .05$ between groups.

Changes in PCS scores were significantly different between groups (main effect: $F_{2,139} = 36.1$, $P < .001$, $\eta_p^2 = .34$; see Fig 6), but the Group × Sex interaction was not statistically significant ($F_{2,139} = .12$, $P = .9$, $\eta_p^2 = .002$). Pain catastrophizing scores were reduced by 8.3 points (95% CI: −10.4 to −6.3) in the SMT group, by 4.9 points (95% CI: −6.9 to −2.9) in the placebo group, and increased by .3 points (95% CI: −1.3 to 1.9) in the control group. Bonferroni-corrected planned contrasts revealed that the reduction in pain catastrophizing scores was greater for the SMT group

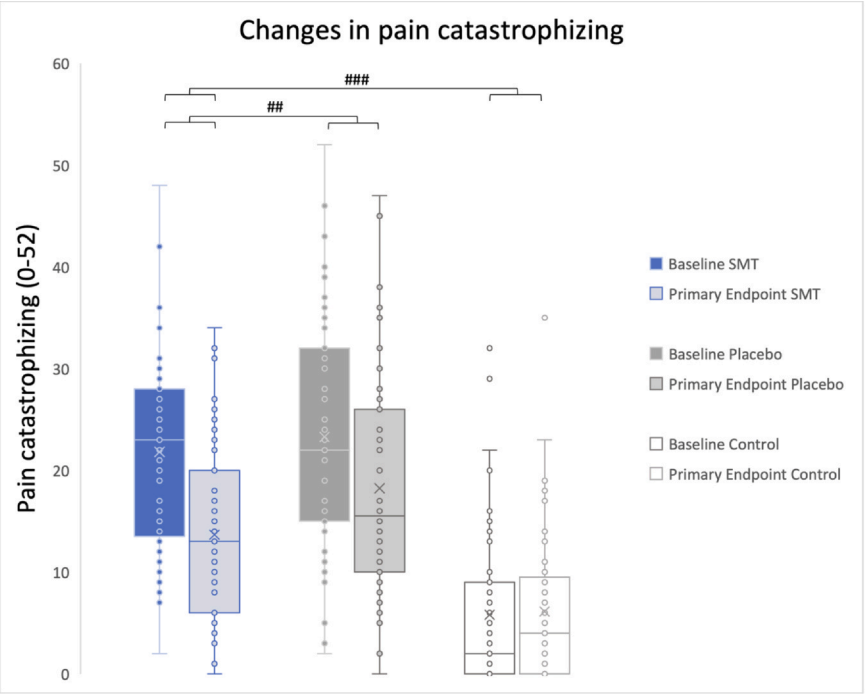


Figure 6. Changes in pain catastrophizing. Boxplots showing the mean, median, and interquartile ranges of pain catastrophizing (0–52) at baseline and at the primary endpoint (4 weeks) for participants in the SMT, placebo, and control groups. The decrease of pain catastrophizing over time was significantly greater in the SMT group compared with the placebo and control groups, and in the placebo group compared with the control group. $^{\#\#}P < .01$, $^{\#\#\#}P < .001$ between groups.

compared with the placebo group (corrected $P = .007$, $d = .51$) and the control group (corrected $P < .001$, $d = 1.22$). No significant difference was observed between the placebo and control groups (corrected $P = .17$, $d = .71$). To rule out that the greater decrease in pain catastrophizing in the SMT group compared with the Placebo group was not confounded by changes in clinical pain intensity,[81,82] the 2 groups were compared with and without pain intensity changes as a covariate using a robust GLM. Without the covariate, changes pain catastrophizing were significantly greater in the SMT group compared with the Placebo group (main effect: $F_{1,92} = 4.9$, $P = .03$, $\eta_p^2 = .05$; mean difference: 3.6 [95% CI, .6–7.1]), but the Group × Sex interaction was not statistically significant ($F_{1,92} = .2$, $P = .7$, $\eta_p^2 = .002$). When accounting for the decrease in clinical pain intensity, changes in pain catastrophizing were not significantly different between the SMT group and the Placebo group (main effect: $F_{1,91} = 2.8$, $P = .096$, $\eta_p^2 = .03$; mean difference: 2.9 [95% CI, –.6 to 6.6]), and the Group × Sex interaction was not statistically significant ($F_{1,91} = .5$, $P = .5$, $\eta_p^2 = .006$). This indicates that the reduction in clinical pain contributed to changes in pain catastrophizing.

Changes in depressive symptoms were significantly different between groups (main effect: $F_{2,139} = 7.0$, $P = .04$, $\eta_p^2 = .09$), but the Group × Sex interaction was not statistically significant ($F_{2,139} = 1.6$, $P = .5$, $\eta_p^2 = .02$). The BDI scores decreased by 2.4 points in the SMT group (95% CI: –3.5 to –1.2) by 2.4 points in the placebo group (95% CI: –3.7 to –1.1) and by .6 points in the control group (95% CI: –1.6 to .4). Bonferroni-corrected planned contrasts revealed that the reduction in depressive symptoms was not significantly different between the SMT group and the placebo group (corrected $P = 1.0$, $d = .03$), between the SMT group and the control group (corrected $P = .09$, $d = .39$), or between the placebo group and the control group (corrected $P = .08$, $d = .36$).

Changes in anxiety were significantly different between groups (main effect: $F_{2,139} = 10.0$, $P = .01$, $\eta_p^2 = .13$), but the Group × Sex interaction was not statistically significant ($F_{2,139} = .5$, $P = .89$, $\eta_p^2 < .01$). Anxiety was reduced by 1.0 point (95% CI: –1.6 to –.3) in the SMT group, by 1.6 points (95% CI: –2.3 to –1.0) in the placebo group, and by .1 points (95% CI: –.7 to .4) in the control group. Bonferroni-corrected planned contrasts revealed that the reduction in anxiety symptoms was not significantly different between the SMT group and the placebo group (corrected $P = .7$, $d = .27$), between the SMT group and the control group (corrected $P = .7$, $d = .36$), or between the placebo group and the control group (corrected $P = .1$, $d = .63$).

In summary, these results indicate that pain catastrophizing was significantly decreased in the SMT group compared with the placebo group, but not central sensitization, depressive symptoms, or anxiety.

Expectations of Pain Relief

Participants in the SMT and placebo groups expected a relatively large, but similar reduction of pain intensity at the primary endpoint (68.8% [95% CI: 62.0–75.6] in the SMT group and 68.7% [95% CI: 60.9–76.5] in the placebo group). Accordingly, expectations of pain reduction were not significantly different between groups ($t = .03$, $P = .97$). It has been shown that expectations of pain relief may contribute to pain reduction over time, with or without an intervention.[74,83] To rule out this effect and to confirm that pain relief by SMT was not the result of expectations, expectations of pain relief were included as a covariate in the statistical model described above (see Efficacy of SMT). This did not change the results and the difference in pain intensity between groups remained significant ($P = .01$). Therefore, expectations did not significantly contribute to the relief of CPLBP by SMT.

Blinding

The certainty of having received SMT was compared between groups over time (after the first, sixth and twelfth treatment sessions) using a mixed ANOVA. Certainty was not significantly different between groups (main effect: $F_{1,92} = .27$, $P = .6$, $\eta_p^2 = .002$), sexes (main effect: $F_{1,92} = .03$, $P = .9$, $\eta_p^2 = .00$), or over time (main effect: $F_{1.9,146} = .06$, $P = .9$, $\eta_p^2 < .01$). Moreover, the Group × Time interaction ($F_{1.6,146} = .77$, $P = .4$, $\eta_p^2 = .002$) and the Group × Sex × Time interaction ($F_{1.6,146} = .34$, $P = .7$, $\eta_p^2 = .00$) were not statistically significant. These findings indicate that patients could not distinguish placebo from SMT until the end of the protocol. For the SMT and placebo groups, certainty was respectively 62.2 (95% CI: 55.5–68.9) and 62.1 (95% CI: 55.2–69.0) after the first session, 61.0 (95% CI: 53.8–68.2) and 64.7 (95% CI: 57.3–72.2) after the sixth session, and 64.6 (95% CI: 56.6–72.5) and 61.9 (95% CI: 53.3–70.6) after the twelfth session. For patients who thought that they had received SMT (SMT group: n = 38; placebo group: n = 34), certainty increased from 66.0 (95% CI: 59.0–73.0) to 71.7 (95% CI: 63.3–80.1), and from 68.2 (95% CI: 60.1–76.3) to 76.7 (95% CI: 70.2–83.2), respectively. For patients who thought that they had not received SMT (SMT group: n = 10; placebo group: n = 14), certainty decreased accordingly from 50.0 (95% CI: 30.5–69.5) to 37.5 (95% CI: 27.2–47.8), and from 48.2 (95% CI: 36.0–60.4) to 26.1 (95% CI: 15.1 –37.1), respectively. To examine if certainty affected pain relief by SMT compared with placebo, ratings of certainty were included as covariates in the statistical model described above (see Efficacy of SMT). Certainty had no significant effect on pain relief by SMT, as the between-group difference in pain intensity remained significant when controlling for any of the 3 certainty ratings ($P = .01$, .006, and .02, respectively). Altogether, these results confirm that blinding was successful and generally, that the SMT and placebo interventions could not be distinguished by participants in either group.

Predictors of the Clinical Response to SMT

The baseline values of the segmental PPTs, the PCS and CSI scores, and expectations of pain relief were assessed as predictors of changes in pain intensity at the primary endpoint using a multiple regression model. No evidence of multicollinearity or autocorrelation among the residuals was found for any of the predictors used in all regression analyses. For the SMT and placebo groups

combined, lower segmental PPTs (higher mechanical sensitivity) significantly predicted greater reductions in CPLBP intensity ($\beta = .22$, $P = .01$). However, the reduction of CPLBP intensity was not predicted significantly by expectations of pain relief ($\beta = .11$, $P = .3$), pain catastrophizing ($\beta = -.20$, $P = .07$), or CSI scores ($\beta = .13$, $P = .3$). In addition, lower baseline segmental PPTs predicted greater reductions in CPLBP intensity in the SMT group ($\beta = .34$, $P = .01$), but not in the placebo group ($\beta = .12$, $P = .3$). Nevertheless, the predictive value of segmental PPTs was not significantly different between the SMT and the placebo groups ($\beta = .26$, $P = .067$).

Despite the lack of group difference in disability over time, regression analyses were conducted to explore potential predictors that may have masked the group effect. For the SMT and placebo groups combined, segmental PPTs, the pain catastrophizing, CSI scores, and expectations of pain relief did not predict changes in disability at the primary endpoint (all P's > .1). In contrast, higher baseline pain catastrophizing predicted the reduction of disability in the SMT group, ($\beta = -.43$, $P = .004$), while no variable predicted reduced disability in the placebo group (all $P > .2$). Nonetheless, the predictive value of pain catastrophizing was not significantly different between the SMT and the placebo groups ($\beta = -.26$, $P = .2$).

Moderators of the Clinical Response to SMT

The moderation of changes in pain intensity by changes in components of nociplastic pain was examined with GLMs. Increases in segmental PPTs significantly moderated the reduction of CPLBP intensity ($\beta = -.24$, $P = .02$, $\eta_p^2 = .06$), but they did not alter the difference between the SMT and placebo groups, which remained significant ($P = .02$). Similarly, reductions in CSI scores significantly moderated the reduction of CPLBP intensity ($\beta = .24$, $P = .01$, $\eta_p^2 = .06$), but they did not alter the difference between the SMT and placebo groups, which remained significant ($P = .003$). Besides, changes in PCS scores and expectations of pain relief did not moderate the reduction of CPLBP intensity ($\beta = .16$, $P = .1$, $\eta_p^2 = .03$ and $\beta = .16$, $P = .1$, $\eta_p^2 = .03$, respectively). This suggests that expectations of pain relief, the increases in segmental PPTs, and the decreases in CSI and PCS scores did not contribute to the difference in pain intensity between the SMT and placebo groups.

Although changes in disability were not significantly different between groups at the primary endpoint, reductions in pain catastrophizing and in CSI scores moderated the decrease in disability ($\beta = .47$, $P < .001$, $\eta_p^2 = .21$ and $\beta = .36$, $P < .001$, $\eta_p^2 = .13$, respectively), but they did not change the lack of group effect (P's = .8 and .5, respectively). Besides, changes in PPTs and expectations did not moderate changes in disability ($\beta = -.10$, $P = .4$, $\eta_p^2 = .009$ and $\beta = .09$, $P = .4$, $\eta_p^2 = .008$, respectively).

Adverse Events

A total of 377 adverse events were reported in the study, 198 in the placebo group and 179 in the SMT group, representing 34.1% and 30.7% of treatment sessions, respectively. The number of events was not significantly different between groups (Mann-Whitney $U = 1{,}261.0$, $P = .7$). Also, no significant difference was observed for the types of events between groups ($\chi^2_{(4,\ 377)} = 4.6$, $P = .3$). The most commonly reported events were an increase in back pain (36.3% in the SMT group, 41.9% in the placebo group), muscle soreness/stiffness (31.3% in the SMT group, 31.8% in the placebo group), pain irradiating to the lower extremity (22.4% in the SMT group, 14.2% in the placebo group), changes in the usual pain pattern or location (7.8% in the SMT group, 9.6% in the placebo group). Finally, some events could not be categorized as one of the above and were labeled "Other" (2.2% in the SMT group and 2.5% in the placebo group).

Regarding the onset of adverse events, it was not significantly different between groups ($\chi^2_{(2,\ 375)} = .2$, $P = .9$). They were reported to begin within 24 hours after the session (52.5% in the SMT group, 53.0% in the placebo group), immediately after the session (28.3% in the SMT group, 29.3% in the placebo group), or > 24 hours after the session (19.2% in the SMT group, 17.7% in the placebo group). Likewise, the duration of adverse events was not significantly different between groups ($\chi^2_{(3,\ 376)} = 1.8$, $P = .6$). They lasted < 24 hours (45.5% in the SMT group, 43.9% in the placebo group), between 24 and 48 hours (28.1% in the SMT group, 24.6% in the placebo group), > 48 hours (16.3% in the SMT group, 18.2% in the placebo group), or < 1 hour (10.1% in the SMT group, 13.3% in the placebo group).

Regarding the severity of adverse events, it was significantly different between groups ($\chi^2_{(4,\ 375)} = 16.4$, $P = .003$). For SMT and placebo groups, events were "Very mild" (23.0% and 26.9%, respectively), Mild (44.4% and 25.4%, respectively), "Moderate" (28.1% and 41.6% respectively), or "Severe" (4.5% and 5.6%, respectively). Only one "Very severe" event was reported by a patient in the placebo group (.5%). All events were transient and self-managed or improved after further care within the 12 sessions, including the very severe event. At the end of the treatment protocol, only 5 participants rated their pain intensity higher compared with baseline. Three rated their pain 10 points higher, one 15 points higher, and one 20 points higher. These 5 participants were all in the placebo group.

Conclusions

Efficacy and Harms of SMT

Previous studies on the efficacy of SMT have suggested that its clinical benefits may rely on nonspecific effects.[22,28,84] In contrast, a clinical trial designed to examine and control for nonspecific effects showed specific pain reduction by SMT.[85] Accordingly, the present study shows that SMT produces greater pain relief compared with a control intervention that was undistinguishable from SMT. This medium effect ($\eta_p^2 = .07$) persisted up to 12 weeks after SMT, suggesting that SMT produces long-lasting pain relief, possibly through specific mechanisms.

Regarding the clinical significance of this effect, pain reduction after 4 weeks of SMT was 30.9 points (65%) lower than baseline. The amplitude of this change is within the range of 2.5 to 4.5 points out of 10, or a 30% pain reduction from baseline, proposed for clinical significance.[86,87] Also, the difference between SMT and placebo (–11.7 points) is superior to the 10-point change proposed for the clinical significance of a placebo-controlled intervention.[88] Although the metric to determine clinical significance is variable,[89,90] the present results indicate that SMT produced a clinically significant reduction of CPLBP.

Consistent with the efficacy of SMT, expectations of pain relief and treatment credibility did not influence between-group differences in pain reduction. Moreover, comparable expectations of pain relief and treatment credibility between groups support the equality assumption between interventions. This suggests that SMT may have produced effects beyond those attributable to placebo.[91] Furthermore, dual blinding limits the potential bias in outcome assessment. Altogether, these factors are consistent with the efficacy of SMT for CPLBP.

Although expectations did not contribute significantly to SMT efficacy, the present results may not generalize to other samples with different expectations of pain relief. Besides, it should be mentioned that although the placebo intervention was not distinguishable from SMT, the 2 interventions are not completely equivalent. Although patients were naïve to chiropractic and SMT, the cavitation and audible release associated with SMT may produce stronger placebo effects compared with the placebo intervention, which generally did not produce cavitation.[92] This is inherent to all placebo-controlled studies on SMT. Therefore, although the present results are statistically and clinically significant, the use of a biomarker specific to SMT, that is not measurable when placebo effects occur, remains to be established to rule out placebo effects completely.

In this regard, previous studies using the same placebo intervention as in the present study reported effects similar to those of SMT.[43,45] In the present study, the placebo group showed a mean pain relief of 17.9 points from baseline, exceeding the estimated short-term effect of 8 points for placebo interventions compared with no intervention.[93] Thus, it could be argued that the placebo intervention also produced clinically meaningful effects.[21] This should be considered in the design of future randomized placebo-controlled trials on SMT, since this placebo intervention does not seem inert and may mask some clinical benefits of SMT. Accordingly, the placebo intervention was associated with a similar frequency, type, and duration of adverse events as SMT, in line with available data.[23,94] Overall, although both interventions may be considered safe, the placebo maneuver was associated with slightly more severe reactions, potentially due to a worse natural progression of CPLBP or to nonspecific effects.[75]

Changes in Mechanical Pain Sensitivity and Pain Catastrophizing

The reduction in segmental mechanical hyperalgesia was significantly greater in the SMT group compared with the placebo group, while changes in other areas did not significantly differ between groups. This suggests that SMT produces segmental effects on mechanical pain sensitivity in a clinical setting. Although the segmental specificity of SMT was questioned recently,[28] the reduction of segmental hyperalgesia may contribute to clinically significant pain relief in patients with CPLBP. This is coherent with previous studies.[21,29,31] One of these studies reported that 4 sessions of SMT reduced mechanical hyperalgesia when SMT targeted the most painful segment.[31] However, no effect was observed on clinical outcomes. As suggested by the authors, the reduction of hyperalgesia may require a longer intervention to translate into clinical benefits.[31,44] This may explain why a statistically and clinically significant reduction of pain intensity was observed with 12 sessions of SMT in the present study.

The present results also imply that SMT does not reduce secondary hyperalgesia. Accordingly, dermatomal (lower limb) mechanical hyperalgesia remained unchanged. In previous studies, it was suggested that spinal manipulation produces immediate effects on secondary mechanical hyperalgesia[29] and on temporal summation of pain in the back,[95] or in the lower limb dermatome corresponding to the manipulated segment, both in healthy volunteers and in individuals with LBP.[85,96,97] Although these results contrast with the present findings, an important methodological difference should be noted. The reported effects in previous studies are immediate and possibly transient effects, while the effects were calculated between baseline and 4 weeks in the present study. Thus, the 2 measures likely capture different neurophysiological processes. In the present study, segmental mechanical hyperalgesia was reduced by SMT in patients with CPLBP, but they still showed segmental mechanical hyperalgesia compared with healthy controls. Greater changes in segmental hyperalgesia were associated with greater changes in secondary hyperalgesia in the present study (not shown). Therefore, it is possible that a greater reduction of segmental hyperalgesia may be necessary to observe significant differences in secondary hyperalgesia.

Patients receiving SMT also showed greater reductions in pain catastrophizing compared with placebo, which was clinically significant.[98] Improvements in pain intensity could contribute to this effect.[81,82] Accordingly, when accounting for greater improvements in pain intensity in the SMT group compared with the Placebo group, the reduction in pain catastrophizing was not statistically significant between groups. This indicates that reductions in clinical pain intensity contribute, at least in part, to the reductions in pain catastrophizing. However, emerging data suggest that manual therapy may reduce pain catastrophizing directly in similar patient populations.[99,100] Although the mechanisms are unclear, SMT could help extinguish conditioned responses between lumbar spine movement and LBP, possibly by influencing brain processes involved in salience detection and pain anticipation.[101,102] This remains to be clarified in future studies.

Moderation of Clinical Improvement by Nociplastic Pain Factors

All nociplastic pain factors assessed in the present study showed a significant alteration in patients with CPLBP compared with controls. Both patient groups showed widespread hyperalgesia, increased pain catastrophizing, central sensitization scores, depressive symptoms, and anxiety. This is consistent with the definition of CPLBP[12,14] and with previous studies.[42,52,103] In the present study, we examined if the reduction of these factors by SMT may contribute to clinical improvement in patients with CPLBP. Only 2 factors, segmental mechanical hyperalgesia and pain catastrophizing, were reduced by SMT. Changes in segmental hyperalgesia moderated pain relief, while changes in pain catastrophizing moderated improvements in disability. In both cases, however, the moderation was nonspecific[104] and did not contribute to the group difference (or lack of). Larger samples may be necessary to detect significant effects.

Previous findings suggest that pain catastrophizing is a predictor of disability[105-107] and a moderator of treatment efficacy, including manual therapy and SMT,[108-112] in individuals with LBP and other musculoskeletal pain. Accordingly, pain catastrophizing contributes to fear-avoidance behaviors and nociplastic pain,[42,113] and its modulation may influence pain sensitivity in healthy individuals and patients with LBP.[114-116] Future research is needed to clarify how pain catastrophizing may moderate the clinical benefits of SMT, or whether SMT reduces pain catastrophizing as a clinical benefit.

Study Limitations and Future Directions

Some limitations should be considered for the interpretation of the present findings. Firstly, although the sample size was large enough to detect significant pain relief by SMT, the between-group difference in disability was not significant. Although average pain intensity was moderate at baseline, patients from both groups displayed relatively low disability compared with previous studies.[117] In addition, the mean ODI scores achieved at all endpoints for both groups fall below the cut-off value for disability.[118] With these levels of disability, the lack of a between-group difference may be explained by a floor effect, which was reported for the ODI.[119]

Secondly, only 1 clinician provided SMT for both groups, which may have introduced a provider or allegiance bias that is inevitable in manual therapy trials.[40,120] Nonetheless, this limitation was mitigated by implementing a successful dual-blind design.[120,121] Additionally, using a single clinician for both SMT and placebo interventions ensures consistency and limits variability between treatments. In future studies, different clinicians could provide different interventions, if this does not compromise feasibility.

In conclusion, the present study indicates that SMT produces greater reductions in clinical pain intensity, pain catastrophizing, and segmental mechanical hyperalgesia compared with a control intervention. These findings are consistent with the improvement of CPLBP by SMT through the regulation of neural processes related to nociplastic pain, although this remains to be confirmed.

Author Contributions

CG-M and **MP** conceptualized and designed the study, except for laboratory analyses, which were conceptualized by **AO-DM**. Data collection was conducted by **BR-S**, **IM-G**, **B-LA**, **CG-M**, and **AO-DM**. Statistical analysis was conducted by **CG-M** and **MP**. **CG-M** and **LA-G** were responsible for ethical committee approval. **CG-M** wrote the first draft of the manuscript. **MP** and **AO-DM** revised the manuscript. **CG-M** and **MP** wrote the final version of the manuscript. All listed authors meet authorship criteria and have read and approved the final manuscript.

Data Availability

All data reported in this study are available from the corresponding author upon reasonable request.

Acknowledgments

The authors thank Dr. Marc Huertas-Company for his help with data analysis, as well as Rafael Marín Cerda, Oscar Mateo Díaz, and Alberto Pérez Tarrazo for their help with data collection.

Disclosures

This work was supported by the Fondation Chiropratique du Québec. The contribution of C.G.-M. was supported by the Fonds de Recherche du Québec en Nature et Technologies (FRQNT), the Asociación Española de Quiropráctica (AEQ), and the European Centre for Chiropractic Research Excellence (ECCRE). The contribution of A.O.-D.M. was supported by the ECCRE. The contribution of M.P. was supported by the Fonds de Recherche du Québec en Santé. Funding sources had no involvement in study design, data collection, analysis, report writing, or the decision to submit for publication. The authors have no financial or other relationship that may lead to a conflict of interest.

Appendix A. Supplementary Data

Supplementary data related to this article can be found at doi:10.1016/j.jpain.2024.02.014.

References

1. G.B.D.Collaborators: Global burden of 369 diseases and injuries in 204 countries and territories, 1990-2019: A systematic analysis for the Global Burden of Disease Study 2019. Lancet 396:1204-1222, 2020.

2. Hartvigsen J, Hancock MJ, Kongsted A, *et al.* What low back pain is and why we need to pay attention. Lancet 391:2356-2367, 2018.

3. Hurwitz EL, Randhawa K, Yu H, Cote P, Haldeman S: The Global Spine Care Initiative: A summary of the global burden of low back and neck pain studies. Eur Spine J: Off Publ Eur Spine Soc Eur Spinal Deformity Soc Eur Section Cervical Spine Res Soc 27:796-801, 2018.

4. Dieleman JL, Cao J, Chapin A, *et al.* US Health Care Spending by Payer and Health Condition, 1996-2016. Jama 323:863-884, 2020.

5. Buchbinder R, Underwood M, Hartvigsen J, Maher CG: The Lancet Series call to action to reduce low value care for low back pain: An update. Pain 161(Suppl 1):S57-S64, 2020.

6. Stevans JM, Delitto A, Khoja SS, *et al.* Risk factors associated with transition from acute to chronic low back pain in US patients seeking primary care. JAMA Netw Open 4:e2037371

7. Kosek E, Cohen M, Baron R, *et al.* Do we need a third mechanistic descriptor for chronic pain states? Pain 157:1382-1386, 2016.

8. Shraim MA, Masse-Alarie H, Hall LM, Hodges PW: Systematic review and synthesis of mechanism-based classification systems for pain experienced in the musculoskeletal system. Clin J Pain 36:793-812, 2020.

9. Fitzcharles MA, Cohen SP, Clauw DJ, Littlejohn G, Usui C, Hauser W: Nociplastic pain: Towards an understanding of prevalent pain conditions. Lancet 397:2098-2110, 2021.

10. Fitzcharles MA, Cohen SP, Clauw DJ, Littlejohn G, Usui C, Hauser W: Chronic primary musculoskeletal pain: A new concept of nonstructural regional pain. Pain Rep 7:e1024

11. Nijs J, Lahousse A, Kapreli E, *et al.* Nociplastic pain criteria or recognition of central sensitization? Pain phenotyping in the past, present and future. J Clin Med 102021.

12. Nicholas MK, Vlaeyen JWS, Rief W, *et al.* The IASP classification of chronic pain for ICD-11: Chronic primary pain. Pain 160:28-37, 2019.

13. Nijs J, Apeldoorn A, Hallegraeff H, *et al.* Low back pain: Guidelines for the clinical classification of predominant neuropathic, nociceptive, or central sensitization pain. Pain Physician 18:E333-E346, 2015.

14. Treede RD, Rief W, Barke A, *et al.* Chronic pain as a symptom or a disease: The IASP Classification of Chronic Pain for the International Classification of Diseases (ICD-11). Pain 160:19-27, 2019.

15. Latremoliere A, Woolf CJ: Central sensitization: A generator of pain hypersensitivity by central neural plasticity. J Pain 10:895-926, 2009.

16. Nijs J, George S, Clauw D, *et al.* Central sensitisation in chronic pain conditions: Latest discoveries and their potential for precision medicine. Lancet Rheumatol 3:e383-e392, 2021.

17. Treede RD, Hoheisel U, Wang D, Magerl W: Central sensitization: Clinical utility of a physiological concept for the International Statistical Classification of Diseases and Related Health Problems and for nociplastic pain. Pain 163:S99-S107, 2022.

18. George SZ, Fritz JM, Silfies SP, *et al.* Interventions for the management of acute and chronic low back pain: Revision 2021. J Orthop Sports Phys Ther 51:CPG1-CPG60, 2021.

19. Qaseem A, Wilt TJ, McLean RM, Forciea MA: Noninvasive treatments for acute, subacute, and chronic low back pain: A clinical practice guideline from the American College of Physicians. Ann Internal Med 166:514-530, 2017.

20. Gevers-Montoro C, Provencher B, Descarreaux M, Ortega de Mues A, Piche M: Clinical effectiveness and efficacy of chiropractic spinal manipulation for spine pain. Front Pain Res 2:765921

21. Gevers-Montoro C, Provencher B, Descarreaux M, Ortega de Mues A, Piche M: Neurophysiological mechanisms of chiropractic spinal manipulation for spine pain. Eur J Pain 25:1429-1448, 2021.

22. Lavazza C, Galli M, Abenavoli A, Maggiani A: Sham treatment effects in manual therapy trials on back pain patients: A systematic review and pairwise meta-analysis. BMJ Open 11:e045106

23. Rubinstein SM, de Zoete A, van Middelkoop M, Assendelft WJJ, de Boer MR, van Tulder MW: Benefits and harms of spinal manipulative therapy for the treatment of chronic low back pain: Systematic review and meta-analysis of randomised controlled trials. BMJ 364:l689, 2019.

24. Bishop F, Al-Abbadey M, Roberts L, *et al.* Direct and mediated effects of treatment context on low back pain outcome: A prospective cohort study. BMJ Open 11:e044831

25. Sherriff B, Clark C, Killingback C, Newell D: Impact of contextual factors on patient outcomes following conservative low back pain treatment: Systematic review. Chiropr Man Ther 30:20, 2022.

26. van Lennep J, Trossel F, Perez R, *et al.* Placebo effects in low back pain: A systematic review and meta-analysis of the literature. Eur J Pain 25:1876-1897, 2021.

27. Cook CE: The demonization of manual therapy. Muskuloskelet Physiother 25:125-132, 2021.

28. Nim CG, Downie A, O'Neill S, Kawchuk GN, Perle SM, Leboeuf-Yde C: The importance of selecting the correct site to apply spinal manipulation when treating spinal pain: Myth or reality? A systematic review. Sci Rep 11:23415, 2021.

29. Gevers-Montoro C, Provencher B, Northon S, Stedile-Lovatel JP, Ortega de Mues A, Piche M: Chiropractic spinal manipulation prevents secondary hyperalgesia induced by topical capsaicin in healthy individuals. Front Pain Res 2:702429

30. Jordon MK, Beattie PF, D'Urso S, Scriven S: Spinal manipulation does not affect pressure pain thresholds in the absence of neuromodulators: A randomized controlled trial. J Man Manip Ther 25:172-181, 2017.

31. Nim CG, Kawchuk GN, Schiottz-Christensen B, O'Neill S: The effect on clinical outcomes when targeting spinal manipulation at stiffness or pain sensitivity: A randomized trial. Sci Rep 10(14615)2020.

32. Provencher B, Northon S, Piche M: Segmental chiropractic spinal manipulation does not reduce pain amplification and the associated pain-related brain activity in a capsaicin-heat pain model. Front Pain Res 2:733727

33. Teodorczyk-Injeyan JA, Triano JJ, Gringmuth R, DeGraauw C, Chow A, Injeyan HS: Effects of spinal manipulative therapy on inflammatory mediators in patients with non-specific low back pain: A non-randomized controlled clinical trial. Chiropr Man Therap 29:3, 2021.

34. Nim CG, Aspinall SL, Weibel R, Steenfelt MG, O'Neill S: Pressure pain thresholds in a real-world chiropractic setting: topography, changes after treatment, and clinical relevance? Chiropr Man Ther 30:25, 2022.

35. Boal RW, Gillette RG: Central neuronal plasticity, low back pain and spinal manipulative therapy. J Manip Physiol Ther 27:314-326, 2004.

36. Zafereo JA, Deschenes BK: The role of spinal manipulation in modifying central sensitization. J Appl Biobehav Res 20:84-99, 2015.

37. Tagliaferri SD, Owen PJ, Miller CT, Mitchell UH, Ehrenbrusthoff K, Belavy DL: Classifying non-specific low back pain for better clinical outcomes: Current challenges and paths forward. J Orthop Sports Phys Ther 53:236-323, 2023.

38. Schulz KF, Altman DG, Moher D: The CONSORT Group: CONSORT 2010 statement: Updated guidelines for reporting parallel group randomised trials. BMJ 340(c332)2010.

39. Howick J, Webster RK, Rees JL, *et al.* TIDieR-Placebo: A guide and checklist for reporting placebo and sham controls. PLoS Med 17:e1003294

40. Hohenschurz-Schmidt D, Vase L, Scott W, *et al.* Recommendations for the development, implementation, and reporting of control interventions in efficacy and mechanistic trials of physical, psychological, and self-management therapies: The CoPPS Statement. BMJ 381:e072108

41. Gevers-Montoro C, Ortega-De Mues A, Piché M: Mechanisms of chiropractic spinal manipulative therapy for patients with chronic primary low back pain: Protocol for a mechanistic randomised placebo-controlled trial. BMJ Open 13:e065999

42. Shraim MA, Masse-Alarie H, Hodges PW: Methods to discriminate between mechanism-based categories of pain experienced in the musculoskeletal system: A systematic review. Pain 162:1007-1037, 2021.

43. Chaibi A, Saltyte Benth J, Bjorn Russell M: Validation of placebo in a manual therapy randomized controlled trial. Sci Rep 5:11774

44. Haas M, Vavrek D, Peterson D, Polissar N, Neradilek MB: Dose-response and efficacy of spinal manipulation for care of chronic low back pain: A randomized controlled trial. Spine J: Off J N Am Spine Soc 14:1106-1116, 2014.

45. Aspinall SL, Jacques A, Leboeuf-Yde C, Etherington SJ, Walker BF: No difference in pressure pain threshold and temporal summation after lumbar spinal manipulation compared to sham: A randomised controlled trial in adults with low back pain. Musculoskelet Sci Pract 43:18-25, 2019.

46. Picchiottino M, Honore M, Leboeuf-Yde C, Gagey O, Cottin F, Hallman DM: The effect of a single spinal manipulation on cardiovascular autonomic activity and the relationship to pressure pain threshold: A randomized, cross-over, sham-controlled trial. Chiropr Man Thera 28:7, 2020.

47. Shraim MA, Sluka KA, Sterling M, *et al.* Features and methods to discriminate between mechanism-based categories of pain experienced in the musculoskeletal system: A Delphi expert consensus study. Pain 163:1812-1828, 2022.

48. Haefeli M, Elfering A: Pain assessment. Eur Spine J: Off Publ Eur Spine Soc Eur Spinal Deformity Soc Eur Section Cervical Spine Res Soc 15(Suppl 1):S17-S24, 2006.

49. Alcántara-Bumbiedro S, Flórez-García M, Echávarri-Pérez C, García-Pérez F: Escala de incapacidad por dolor lumbar de Oswestry. Rehabilitación 40:150-158, 2006.

50. Arendt-Nielsen L, Morlion B, Perrot S, *et al.* Assessment and manifestation of central sensitisation across different chronic pain conditions. Eur J Pain 22:216-241, 2018.

51. Fischer AA: Pressure threshold measurement for diagnosis of myofascial pain and evaluation of treatment results. Clin J Pain 2:207-214, 1986.

52. den Bandt HL, Paulis WD, Beckwee D, Ickmans K, Nijs J, Voogt L: Pain mechanisms in low back pain: A systematic review with meta-analysis of mechanical quantitative sensory testing outcomes in people with nonspecific low back pain. J Orthop Sports Phys Ther 49:698-715, 2019.

53. Neziri AY, Curatolo M, Limacher A, *et al.* Ranking of parameters of pain hypersensitivity according to their discriminative ability in chronic low back pain. Pain 153:2083-2091, 2012.

54. Vuilleumier PH, Biurrun Manresa JA, Ghamri Y, *et al.* Reliability of quantitative sensory tests in a low back pain population. Reg Anesth Pain Med 40:665-673, 2015.

55. Pfau DB, Krumova EK, Treede RD, *et al.* Quantitative sensory testing in the German Research Network on Neuropathic Pain (DFNS): Reference data for the trunk and application in patients with chronic postherpetic neuralgia. Pain 155:1002-1015, 2014.

56. Garcia Campayo J, Rodero B, Alda M, Sobradiel N, Montero J, Moreno S: Validation of the Spanish version of the Pain Catastrophizing Scale in fibromyalgia. Med Clin 131:487-492, 2008.

57. Cuesta-Vargas AI, Roldan-Jimenez C, Neblett R, Gatchel RJ: Cross-cultural adaptation and validity of the Spanish central sensitization inventory. Springerplus 5(1837)2016.

58. Holm LA, Nim CG, Lauridsen HH, Filtenborg JB, O'Neill SF: Convergent validity of the central sensitization inventory and experimental testing of pain sensitivity. Scand J Pain 22:597-613, 2022.

59. Huysmans E, Ickmans K, Van Dyck D, *et al.* Association between symptoms of central sensitization and cognitive behavioral factors in people with chronic nonspecific low back pain: A cross-sectional study. J Manip Physiol Ther 41:92-101, 2018.

60. Roussel NA, Nijs J, Meeus M, Mylius V, Fayt C, Oostendorp R: Central sensitization and altered central pain processing in chronic low back pain: Fact or myth? Clin J Pain 29:625-638, 2013.

61. Scerbo T, Colasurdo J, Dunn S, Unger J, Nijs J, Cook C: Measurement properties of the central sensitization inventory: A systematic review. Pain Pract 18:544-554, 2018.

62. Edwards RR, Dworkin RH, Sullivan MD, Turk DC, Wasan AD: The role of psychosocial processes in the development and maintenance of chronic pain. J Pain 17:T70-92, 2016.

63. Owens MA, Bulls HW, Trost Z, *et al.* An examination of pain catastrophizing and endogenous pain modulatory processes in adults with chronic low back pain. Pain Med 17:1452-1464, 2016.

64. García-Campayo J, Zamorano E, Ruiz MA, *et al.* Cultural adaptation into Spanish of the generalized anxiety disorder-7 (GAD-7) scale as a screening tool. Health Qual Life Outcomes 8:8, 2010.

65. Sanz J, García-Vera MP, Espinosa R, Fortún M, Vázquez C: Adaptación española del Inventario para la Depresión de Beck-II (BDI-II): 3. Propiedades psicométricas en pacientes con trastornos psicológicos. Clín Salud 16:121-142, 2005.

66. Gore M, Sadosky A, Stacey BR, Tai KS, Leslie D: The burden of chronic low back pain: Clinical comorbidities, treatment patterns, and health care costs in usual care settings. Spine 37:E668-E677, 2012.

67. Wong JJ, Tricco AC, Cote P, *et al.* Association between depressive symptoms or depression and health outcomes for low back pain: A systematic review and meta-analysis. J Gen Intern Med 37:1233-1246, 2021.

68. Aoyagi K, He J, Nicol AL, *et al.* A subgroup of chronic low back pain patients with central sensitization. Clin J Pain 35:869-879, 2019.

69. Clark JR, Nijs J, Yeowell G, Holmes P, Goodwin PC: Trait sensitivity, anxiety, and personality are predictive of central sensitization symptoms in patients with chronic low back pain. Pain Pract 19:800-810, 2019.

70. Smart KM, Blake C, Staines A, Thacker M, Doody C: Mechanisms-based classifications of musculoskeletal pain: Part 1 of 3: symptoms and signs of central sensitisation in patients with low back (+/- leg) pain. Man Ther 17:336-344, 2012.

71. Kongsted A, Hestbaek L, Kent P: How can latent trajectories of back pain be translated into defined subgroups? BMC Musculoskelet Disord 18:285, 2017.

72. Kosek E, Clauw D, Nijs J, *et al.* Chronic nociplastic pain affecting the musculoskeletal system: Clinical criteria and grading system. Pain 162:2629-2634, 2021.

73. Langford DJ, Lou R, Sheen S, *et al.* Expectations for improvement: A neglected but potentially important covariate or moderator for chronic pain clinical trials. J Pain 24:575-581, 2022.

74. Cormier S, Lavigne GL, Choiniere M, Rainville P: Expectations predict chronic pain treatment outcomes. Pain 157:329-338, 2016.

75. Walker BF, Hebert JJ, Stomski NJ, *et al.* Outcomes of usual chiropractic. The OUCH randomized controlled trial of adverse events. Spine 38:1723-1729, 2013.

76. Ortega Calvo M, Cayuela Dominguez A: Unconditioned logistic regression and sample size: A bibliographic review. Rev Esp Salud Publica 76:85-93, 2002.

77. Mair P, Wilcox R: Robust statistical methods in R using the WRS2 package. Behav Res Methods 52:464-488, 2020.

78. Mooney CZ, Mooney CF, Duval RD, Mooney CL, Duvall R: Bootstrapping: A Nonparametric Approach to Statistical Inference. Sage; 1993

79. Dong Y, Peng CY: Principled missing data methods for researchers. Springerplus 2:222, 2013.

80. Schafer JL: Multiple imputation: A primer. Stat Methods Med Res 8:3-15, 1999.

81. Hirata J, Tomiyama M, Koike Y, Yoshimura M, Inoue K: Relationship between pain intensity, pain catastrophizing, and self-efficacy in patients with frozen shoulder: A cross-sectional study. J Orthop Surg Res 16:542, 2021.

82. Suso-Ribera C, Garcia-Palacios A, Botella C, Ribera-Canudas MV: Pain catastrophizing and its relationship with health outcomes: Does pain intensity matter? Pain Res Manag 2017:9762864, 2017.

83. Bishop MD, Bialosky JE, Cleland JA: Patient expectations of benefit from common interventions for low back pain and effects on outcome: Secondary analysis of a clinical trial of manual therapy interventions. J Man Manip Ther 19:20-25, 2011.

84. Newell D, Lothe LR, Raven TJL: Contextually Aided Recovery (CARe): A scientific theory for innate healing. Chiropr Man Therap 25:6, 2017.

85. Bialosky JE, George SZ, Horn ME, Price DD, Staud R, Robinson ME: Spinal manipulative therapy-specific changes in pain sensitivity in individuals with low back pain (NCT01168999). J Pain 15:136-148, 2014.

86. Ostelo RW, Deyo RA, Stratford P, *et al.* Interpreting change scores for pain and functional status in low back pain: Towards international consensus regarding minimal important change. Spine 33:90-94, 2008.

87. van der Roer N, Ostelo RW, Bekkering GE, van Tulder MW, de Vet HC: Minimal clinically important change for pain intensity, functional status, and general health status in patients with nonspecific low back pain. Spine 31:578-582, 2006.

88. Busse JW, Bartlett SJ, Dougados M, *et al.* Optimal strategies for reporting pain in clinical trials and systematic reviews: Recommendations from an OMERACT 12 Workshop. J Rheumatol 42:1962-1970, 2015.

89. Cook CE, Bejarano G, Reneker J, Vigotsky AD, Riddle DL: Responder analyses: A methodological mess. J Orthop Sports Phys Ther 53:652-654, 2023.

90. Franceschini M, Boffa A, Pignotti E, Andriolo L, Zaffagnini S, Filardo G: The minimal clinically important difference changes greatly based on the different calculation methods. Am J Sports Med 51:1067-1073, 2023.

91. D'Alessandro G, Ruffini N, Aquino A, *et al.* Differences between experimental and placebo arms in manual therapy trials: A methodological review. BMC Med Res Methodol 22:1-14, 2022.

92. Moorman AC, Newell D: Impact of audible pops associated with spinal manipulation on perceived pain: A systematic review. Chiropr Man Therap 30:42, 2022.

93. Strijkers RHW, Schreijenberg M, Gerger H, Koes BW, Chiarotto A: Effectiveness of placebo interventions for patients with nonspecific low back pain: A systematic review and meta-analysis. Pain 162:2792-2804, 2021.

94. Swait G, Finch R: What are the risks of manual treatment of the spine? A scoping review for clinicians. Chiropr Man Therap 25:37, 2017.

95. Randoll C, Gagnon-Normandin V, Tessier J, *et al.* The mechanism of back pain relief by spinal manipulation relies on decreased temporal summation of pain. Neuroscience 349:220-228, 2017.

96. Bialosky JE, Bishop MD, Robinson ME, Zeppieri Jr. G, George SZ: Spinal manipulative therapy has an immediate effect on thermal pain sensitivity in people with low back pain: A randomized controlled trial. Phys Ther 89:1292-1303, 2009.

97. George SZ, Bishop MD, Bialosky JE, Zeppieri Jr. G, Robinson ME: Immediate effects of spinal manipulation on thermal pain sensitivity: An experimental study. BMC Musculoskelet Disord 7:68-78, 2006.

98. Suzuki H, Aono S, Inoue S, *et al.* Clinically significant changes in pain along the Pain Intensity Numerical Rating Scale in patients with chronic low back pain. PloS One 15:e0229228

99. Kamonseki DH, Christenson P, Rezvanifar SC, Calixtre LB: Effects of manual therapy on fear avoidance, kinesiophobia and pain catastrophizing in individuals with chronic musculoskeletal pain: Systematic review and meta-analysis. Musculoskelet Sci Pract 51:102311

100. Kim H, Lee S: The impact of manual therapy on pain catastrophizing in chronic pain conditions: A systematic review and meta-analysis. Phys Ther Rehabilit Sci 12:177-184, 2023.

101. Ellingsen DM, Napadow V, Protsenko E, *et al.* Brain mechanisms of anticipated painful movements and their modulation by manual therapy in chronic low back pain. J Pain 19:1352-1365, 2018.

102. Isenburg K, Mawla I, Loggia ML, *et al.* Increased salience network connectivity following manual therapy is associated with reduced pain in chronic low back pain patients. J Pain 22:545-555, 2021.

103. den Bandt HL, Ickmans K, Leemans L, Nijs J, Voogt L: Differences in quantitative sensory testing outcomes between patients with low back pain in primary care and pain-free controls. Clin J Pain 38:381-387, 2022.

104. Kraemer HC, Wilson GT, Fairburn CG, Agras WS: Mediators and moderators of treatment effects in randomized clinical trials. Arch Gen Psychiatry 59:877-883, 2002.

105. Kovacs FM, Seco J, Royuela A, Pena A, Muriel A: Spanish Back Pain Research Network: The correlation between pain, catastrophizing, and disability in subacute and chronic low back pain: A study in the routine clinical practice of the Spanish National Health Service. Spine 36:339-345, 2011.

106. Martinez-Calderon J, Jensen MP, Morales-Asencio JM, Luque-Suarez A: Pain catastrophizing and function in individuals with chronic musculoskeletal pain: A systematic review and meta-analysis. Clin J Pain 35:279-293, 2019.

107. Wertli MM, Eugster R, Held U, Steurer J, Kofmehl R, Weiser S: Catastrophizing-a prognostic factor for outcome in patients with low back pain: A systematic review. Spine J: Off J N Am Spine Soc 14:2639-2657, 2014.

108. Alonso-Perez JL, Lopez-Lopez A, La Touche R, *et al.* Hypoalgesic effects of three different manual therapy techniques on cervical spine and psychological interaction: A randomized clinical trial. J Bodywork Mov Ther 21:798-803, 2017.

109. Bishop MD, Torres-Cueco R, Gay CW, Lluch-Girbes E, Beneciuk JM, Bialosky JE: What effect can manual therapy have on a patient's pain experience? Pain Manag 5:455-464, 2015.

110. Gevers-Montoro C, Deldar Z, Conesa-Buendia FM, *et al.* Pain catastrophizing mediates rapid benefits of accessing in-person chiropractic care during the COVID-19 lockdown. Eur J Pain 26:463-479, 2021.

111. Verhagen AP, Karels CH, Schellingerhout JM, Willemsen SP, Koes BW, Bierma-Zeinstra SM: Pain severity and catastrophising modify treatment success in neck pain patients in primary care. Man Ther 15:267-272, 2010.

112. Wertli MM, Burgstaller JM, Weiser S, Steurer J, Kofmehl R, Held U: Influence of catastrophizing on treatment outcome in patients with nonspecific low back pain: A systematic review. Spine 39:263-273, 2014.

113. Quartana PJ, Campbell CM, Edwards RR: Pain catastrophizing: A critical review. Expert Rev Neurother 9:745-758, 2009.

114. Meints SM, Mawla I, Napadow V, *et al.* The relationship between catastrophizing and altered pain sensitivity in patients with chronic low-back pain. Pain 160:833-843, 2019.

115. Salomons TV, Moayedi M, Erpelding N, Davis KD: A brief cognitive-behavioural intervention for pain reduces secondary hyperalgesia. Pain 155:1446-1452, 2014.

116. Taub CJ, Sturgeon JA, Johnson KA, Mackey SC, Darnall BD: Effects of a pain catastrophizing induction on sensory testing in women with chronic low back pain: A pilot study. Pain Res Manag 2017:7892494, 2017.

117. Fairbank JC, Pynsent PB: The Oswestry Disability Index. Spine 25:2940-2952, 2000.

118. Tonosu J, Takeshita K, Hara N, *et al.* The normative score and the cut-off value of the Oswestry Disability Index (ODI). Eur Spine J: Off Publ Eur Spine Soc Europ Spinal Deformity Soc Eur Section Cervical Spine Res Soc 21:1596-1602, 2012.

119. Brodke DS, Goz V, Lawrence BD, Spiker WR, Neese A, Hung M: Oswestry Disability Index: A psychometric analysis with 1,610 patients. Spine J: Off J N Am Spine Soc 17:321-327, 2017.

120. D'Alessandro G, Ruffini N, Iacopini A, Annoni M, Kossowsky J, Cerritelli F: Five challenges for manual therapies trials with placebo controls: A proposal. Int J Osteopathic Med 46:55-59, 2022.

121. Caspi O, Millen C, Sechrest L: Integrity and research: introducing the concept of dual blindness: How blind are double-blind clinical trials in alternative medicine? J Altern Complement Med 6:493-498, 2000.